



**COMSATS University Islamabad,
Abbottabad Campus**

**SOFTWARE REQUIREMENTS
SPECIFICATION
(SRS DOCUMENT)**

for

INTELI-REHAB

A Smart Wearable Rehabilitation System

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken
Comments for 10%: Task list for 30%: Verbally communicated as mentioned in presentation.	We have implemented UI for all three actors (Patient, Physiotherapist and Clinic) with user authentication and constructed the wearable prototype.

Supervised by
Dr. Ahmad Khan

Signature_____

Introduction

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements of the IntelliRehab system, a smart wearable rehabilitation platform that combines sensor-based movement tracking, EMG analysis, and 3D digital twin visualization. This document provides a structured overview of the system requirements and serves as a reference for developers, testers, physiotherapists, and other stakeholders involved in the design, development, and evaluation of the system.

Purpose

This Software Requirements Specification (SRS) document defines the requirements of the IntelliRehab system, a smart rehabilitation platform that uses wearable sensors, AI-based analysis, and 3D visualization to monitor and improve patient recovery exercises. The purpose of this document is to provide a clear description of the system's functionalities, constraints, and expected behavior. It will serve as a reference for developers, testers, physiotherapists, and stakeholders throughout the design, development, and validation phases of the project.

Scope

In Scope

1. Wearable hardware prototype with dual IMU (MPU6500) and dual-channel EMG sensors, BLE-enabled microcontroller
2. Flutter-based Android mobile application for BLE connectivity, sensor data reception, calibration, and exercise session management
3. Sensor fusion for joint angles and ROM computation
4. EMG signal processing: filtering, feature extraction, and fatigue trend analysis
5. 3D digital twin visualization (Unity/Three.js) with real-time movement rendering and color-coded deviation feedback
6. AI anomaly detection for abnormal patterns, compensatory behaviours, and early fatigue alerts
7. Gamification: repetition counting, performance scoring, badges, and achievement levels
8. Supabase cloud backend for authentication, encrypted storage, session sync, and multi-device support
9. Web based physiotherapist dashboard for remote monitoring, exercise approval, protocol adjustments, and report generation and a clinic web panel to appoint physiotherapists
10. Upper-limb musculoskeletal rehabilitation exercises under physiotherapist supervision

Out of Scope

1. Live video-based pose estimation systems
2. Clinical diagnostic capabilities beyond exercise guidance
3. Integration with hospital electronic medical record (EMR) systems
4. Physics-based biomechanical force simulations
5. Medical prescriptions or replacement of professional clinical diagnosis

Overall description

1. Product perspective

- IntelliRehab is a new standalone rehabilitation system developed as a Final Year Project (FYP) and is not an extension or replacement of any existing product.
- The system is designed to provide a smart rehabilitation solution by combining wearable sensors, AI-based analysis, and 3D digital twin visualization.
- It consists of a wearable armband with IMU and EMG sensors, a mobile application, and a backend platform for managing rehabilitation data.
- The system enables patients to perform guided exercises while allowing physiotherapists to monitor progress and evaluate recovery.
- The application integrates real-time movement tracking, muscle activity monitoring, and exercise performance assessment.
- The architecture supports future enhancements such as advanced AI models, additional rehabilitation exercises, and healthcare system integration.

2. Operating environment

- The system shall operate on Android-based mobile devices with a minimum OS requirement of Android 10 (API Level 29) or higher to support smooth application performance.
- The mobile application shall be developed using Flutter framework and shall communicate with the backend services for data processing, user management, and rehabilitation record handling.
- The system shall use a cloud-based backend with PostgreSQL database for secure storage and management of patient, physiotherapist, exercise, and sensor data.
- The wearable device shall operate using an ESP32-based microcontroller connected with IMU and EMG sensors to collect real-time movement and muscle activity data.
- The system shall require Bluetooth Low Energy (BLE) connectivity for communication between the wearable sensors and the mobile application.

- An active internet connection shall be required for user authentication, cloud synchronization, progress tracking, and physiotherapist monitoring features.
- The physiotherapist dashboard shall be accessible through an authenticated interface connected to the same backend environment for managing patients and reviewing rehabilitation progress.
- The system shall use secure cloud infrastructure to ensure data availability, backup, and reliable storage of rehabilitation records.

3. Design and implementation constraints

- CO-1: The system shall be developed using the Flutter framework for mobile application development to ensure cross-platform compatibility, efficient UI development, and consistent performance.
- CO-2: The backend services shall use a cloud-based architecture with PostgreSQL database support to provide secure data storage, scalability, and efficient management of rehabilitation records.
- CO-3: The wearable device shall be implemented using an ESP32-based microcontroller with IMU and EMG sensors due to its low power consumption, BLE support, and suitability for real-time sensor data transmission.
- CO-4: The system shall require Bluetooth Low Energy (BLE) connectivity between the wearable device and mobile application, limiting operation to devices that support BLE communication.
- CO-5: The system shall require an active internet connection for cloud synchronization, patient data management, and physiotherapist monitoring features.
- CO-6: The AI-based analysis shall rely on available rehabilitation datasets and predefined models due to limitations in collecting large-scale clinical data during the project development phase.
- CO-7: The 3D digital twin visualization shall be optimized for mobile performance, considering device processing capabilities and memory limitations.

Requirement identifying technique

1. Use case diagram



Figure 1 Use-Case Diagram

2. Use Case Description

A fully dressed use case is a detailed, comprehensive specification that describes a complete interaction between an actor and the system, including preconditions, normal flow, alternative flows, postconditions, exceptions, business rules, and assumptions.

Table 1 UC-1: Register/Login

Use Case ID	UC-1
Use Case Name	Register / Login
Actors	Primary Actor: Patient, Physiotherapist, Clinic Administrator
Description	This use case allows a new patient, physiotherapist, or clinic administrator to register and allows existing users to log into the Inteli-Rehab system using secure credentials. Successful authentication grants access to role-specific dashboards and features.
Trigger	The user opens the Inteli-Rehab application and selects Register or Login.
Preconditions	• PRE-1: User must have an active internet connection. • PRE-2: Authentication service and database must be operational. • PRE-3: Existing users must already have registered accounts.
Postconditions	• POST-1: User account is created successfully (for registration). • POST-2: User is authenticated successfully. • POST-3: User is redirected to the appropriate dashboard according to their role. • POST-4: Secure authentication token is generated.
Normal Flow	1. The user selects Register or Login. 2. The system displays the corresponding form. 3. The user enters the required credentials and personal information.

	4. The system validates all input fields. 5. The system verifies the credentials (or creates a new account). 6. Authentication succeeds. 7. The system generates a secure session. 8. The user is redirected to the appropriate dashboard.
Alternative Flows	1.1 Existing Email1. The system detects that the email already exists during registration.2. The user is prompted to log in instead. 1.2 Invalid Credentials1. The entered email or password is incorrect. 2. The system requests the user to re-enter valid credentials.
Exceptions	• E-1: Authentication server unavailable. • E-2: Internet connection lost during authentication.
Business Rules	• BR-1: Each email address must be unique. • BR-2: Password must contain at least 8 characters including letters, numbers, and a special character. • BR-3: User access is determined according to assigned role.
Assumptions	• A-1: User has a valid email address. • A-2: Authentication service is properly configured.

Table 2 UC-2: Select Clinic

Use Case ID	UC-2
Use Case Name	Select Clinic
Actors	Primary Actor: Patient

Description	This use-case allows the patient to select a registered clinic from the available list. The selected clinic determines the physiotherapists and rehabilitation programs available to the patient.
Trigger	The patient completes registration/login and selects Choose Clinic.
Preconditions	• PRE-1: Patient must be signing up. • PRE-2: At least one clinic must be registered in the system. • PRE-3: Clinic database must be accessible.
Postconditions	• POST-1: Selected clinic is saved in the patient's profile. • POST-2: Associated physiotherapists become available for selection.
Normal Flow	1. The patient selects Choose Clinic. 2. The system retrieves all registered clinics. 3. The clinics are displayed. 4. The patient selects a preferred clinic. 5. The system validates the selection. 6. The selected clinic is saved. 7. The system displays available physiotherapists from the selected clinic.
Alternative Flows	1.1 Search Clinic1. Patient searches by clinic name.2. The system filters matching clinics. 1.2 Change Clinic1. Patient changes clinic before selecting a physiotherapist.2. Previous selection is replaced.
Exceptions	• E-1: Clinic database unavailable. • E-2: No clinics available.
Business Rules	• BR-1: Patient can only select one clinic at a time. • BR-2: Only verified clinics appear in the list.

Assumptions	• A-1: Clinics have already been registered by the administrator.
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Table 3 UC-3: Select Physiotherapist

Use Case ID	UC-3
Use Case Name	Select Physiotherapist
Actors	Primary Actor: Patient
Description	This use-case enables the patient to choose a physiotherapist from the selected clinic. The selected physiotherapist becomes responsible for monitoring the patient's rehabilitation and assigning exercise plans.
Trigger	The patient selects Choose Physiotherapist after selecting a clinic.
Preconditions	• PRE-1: Patient must be logged in. • PRE-2: A clinic must already be selected. • PRE-3: Physiotherapists must be available in the selected clinic.
Postconditions	• POST-1: Selected physiotherapist is assigned to the patient. • POST-2: Patient can receive personalized rehabilitation plans.
Normal Flow	1. The patient selects Choose Physiotherapist. 2. The system retrieves physiotherapists associated with the selected clinic. 3. The available physiotherapists are displayed. 4. The patient reviews profiles. 5. The patient selects a physiotherapist. 6. The system stores the assignment.

	7. Confirmation message is displayed.
Alternative Flows	1.1 Change Physiotherapist 1. Patient selects another physiotherapist before rehabilitation begins. 2. The assignment is updated. 1.2 No Physiotherapist Available 1. The system informs the patient that no physiotherapist is currently available.
Exceptions	• E-1: Server error while retrieving physiotherapists. • E-2: Database connection failure.
Business Rules	• BR-1: Patient can have only one assigned physiotherapist at a time. • BR-2: Only physiotherapists belonging to the selected clinic can be assigned.
Assumptions	• A-1: Physiotherapist accounts have already been added by the clinic.

Table 4 UC-4: Connect Wearable Device

Use Case ID	UC-4
Use Case Name	Connect Wearable Device
Actors	Primary Actor: Patient Secondary Actor: Wearable Sensor Armband
Description	This use-case allows the patient to connect the wearable rehabilitation armband to the mobile application through Bluetooth Low Energy (BLE). Once connected, the wearable continuously transmits IMU and EMG sensor data during rehabilitation sessions.
Trigger	The patient selects Connect Wearable Device.
Preconditions	• PRE-1: Patient must be logged in/signing up. • PRE-2: Bluetooth must be enabled.

	• PRE-3: Wearable armband must be powered on and within Bluetooth range.
Postconditions	• POST-1: BLE connection is established. • POST-2: Sensor data streaming becomes available. • POST-3: Device status changes to Connected.
Normal Flow	1. Patient selects Connect Wearable Device. 2. The system scans for nearby BLE devices. 3. The wearable armband is detected. 4. Patient selects the detected device. 5. Secure BLE pairing is established. 6. Device information is verified. 7. Sensor streaming begins. 8. Connection success message is displayed.
Alternative Flows	1.1 Device Not Found 1. System continues scanning. 2. User retries scanning. 1.2 Reconnect Device 1. Previously paired device reconnected automatically.
Exceptions	• E-1: Bluetooth disabled. • E-2: BLE pairing failure. • E-3: Device moves out of communication range.
Business Rules	• BR-1: Only authorized Intelli-Rehab wearable devices may connect. • BR-2: Only one wearable device may remain connected per rehabilitation session.
Assumptions	• A-1: Wearable firmware is functioning correctly. • A-2: BLE communication is supported by the mobile device.

Table 5 UC-5: Calibrate Sensors

Use Case ID	UC-5
Use Case Name	Calibrate Sensors
Actors	Primary Actor: Patient Secondary Actor: Wearable Sensor Armband
Description	This use case establishes the patient's personalized biomechanical baseline by recording the resting arm position and baseline muscle activity. The calibration ensures accurate movement tracking and personalized rehabilitation analysis throughout future exercise sessions.
Trigger	The patient selects Calibrate Sensors after connecting the wearable device.
Preconditions	• PRE-1: Wearable device must be connected successfully. • PRE-2: Patient must wear the armband correctly. • PRE-3: Sensor communication must be active.
Postconditions	• POST-1: Baseline IMU orientation is stored. • POST-2: Baseline EMG values are recorded. • POST-3: Patient is ready to begin rehabilitation sessions.
Normal Flow	1. Patient selects Calibrate Sensors. 2. The system displays calibration instructions. 3. Patient keeps the arm in the neutral resting position. 4. The wearable collects IMU and EMG readings. 5. The system computes baseline joint orientation and muscle activity. 6. Calibration values are validated. 7. The baseline profile is stored.

	8. Calibration successful message is displayed.
Alternative Flows	1.1 Repeat Calibration 1. Calibration accuracy is insufficient. 2. The system requests another calibration attempt. 1.2 Armband Repositioned 1. Patient adjusts the wearable placement. 2. Calibration restarts automatically.
Exceptions	• E-1: Sensor data unavailable. • E-2: Wearable disconnects during calibration. • E-3: Calibration timeout due to excessive movement.
Business Rules	• BR-1: Calibration must be completed before any rehabilitation session starts. • BR-2: Recalibration is required whenever the wearable position changes significantly.
Assumptions	• A-1: Patient follows the displayed calibration instructions. • A-2: Sensors are functioning within acceptable accuracy limits.

Table 6 UC-6: View Assigned Exercise Plan

Use Case ID	UC-6
Use Case Name	View Assigned Exercise Plan
Actors	Primary Actor: Patient
Description	This use-case allows the patient to view the personalized rehabilitation exercise plan assigned by the physiotherapist. The plan includes exercise instructions, repetitions, sets, session frequency, ROM targets, and rehabilitation goals.
Trigger	The patient selects View Assigned Exercise Plan from the dashboard.
Preconditions	• PRE-1: Patient must be logged in.

	• PRE-2: A physiotherapist must be assigned. • PRE-3: At least one rehabilitation plan must be assigned to the patient.
Postconditions	• POST-1: Assigned exercise plan is displayed. • POST-2: Patient can start the rehabilitation session from the displayed plan.
Normal Flow	1. The patient selects View Assigned Exercise Plan. 2. The system retrieves the latest exercise plan from the database. 3. The exercise list is displayed. 4. The patient reviews exercise descriptions, repetitions, sets, and ROM targets. 5. The patient selects an exercise session. 6. The system enables the Start Rehabilitation Session option.
Alternative Flows	1.1 Updated Exercise Plan1. The physiotherapist has modified the plan. 2. The latest version is automatically displayed. 1.2 No Exercise Assigned1. The system informs the patient that no exercise plan has been assigned yet.
Exceptions	• E-1: Database connection failure. • E-2: Exercise plan retrieval timeout.
Business Rules	• BR-1: Patients can only view exercise plans assigned by their own physiotherapist. • BR-2: Patients cannot modify exercise plans.
Assumptions	• A-1: Physiotherapist has already assigned the rehabilitation protocol.

Table 7 UC-7: Start Rehabilitation Session

Use Case ID	UC-7
Use Case Name	Start Rehabilitation Session
Actors	Primary Actor: Patient Secondary Actor: Wearable Sensor Armband, AI Analysis Engine
Description	This use case allows the patient to begin a rehabilitation session. The system verifies wearable connectivity, calibration status, and loads the assigned exercise protocol before collecting live sensor data throughout the session.
Trigger	The patient selects Start Rehabilitation Session.
Preconditions	• PRE-1: Patient must be logged in. • PRE-2: Wearable device must be connected. • PRE-3: Sensor calibration must be completed. • PRE-4: Exercise plan must be assigned.
Postconditions	• POST-1: Rehabilitation session begins successfully. • POST-2: Live sensor data collection starts. • POST-3: Session activity is recorded for later review.
Normal Flow	1. The patient selects Start Rehabilitation Session. 2. The system verifies BLE connectivity. 3. The system checks calibration status. 4. The assigned exercise plan is loaded. 5. Live IMU and EMG data collection begins.

	6. Digital Twin and AI modules are activated. 7. The rehabilitation session starts.
Alternative Flows	1.1 Calibration Required 1. Calibration is outdated. 2. The system redirects the patient to sensor calibration. 1.2 Device Disconnected 1. The wearable connection is lost. 2. The system requests reconnection before starting.
Exceptions	• E-1: BLE communication failure. • E-2: Sensor initialization failure. • E-3: Exercise plan unavailable.
Business Rules	• BR-1: Rehabilitation sessions cannot begin without successful calibration. • BR-2: Only one rehabilitation session may be active at a time.
Assumptions	• A-1: Wearable device battery is sufficiently charged. • A-2: Patient follows physiotherapist instructions.

Table 8 UC-8: Perform Exercise

Use Case ID	UC-8
Use Case Name	Perform Exercise
Actors	Primary Actor: Patient Secondary Actor: Wearable Sensor Armband, AI Analysis Engine
Description	This use-case enables the patient to perform rehabilitation exercises while the system continuously tracks joint movement, muscle activation, repetition count, and exercise quality using IMU and EMG sensors.
Trigger	The rehabilitation session begins and the patient starts performing an assigned exercise.

Preconditions	• PRE-1: Rehabilitation session must be active. • PRE-2: Wearable sensors must be connected and calibrated. • PRE-3: Assigned exercise instructions must be loaded.
Postconditions	• POST-1: Exercise performance data is recorded. • POST-2: Repetitions and ROM measurements are updated. • POST-3: Exercise statistics are stored in the database.
Normal Flow	1. The system displays exercise instructions. 2. The patient performs the exercise. 3. IMU and EMG sensors collect movement and muscle activity. 4. The system calculates joint angles and ROM. 5. Completed repetitions are counted automatically. 6. Performance metrics are continuously updated. 7. Exercise completion is recorded.
Alternative Flows	1.1 Exercise Paused 1. Patient pauses the exercise. 2. The system temporarily suspends data recording. 1.2 Exercise Completed Early 1. Patient finishes before target repetitions. 2. Session is recorded with completed repetitions only.
Exceptions	• E-1: Sensor malfunction during exercise. • E-2: Unexpected application crash. • E-3: Bluetooth communication interruption.
Business Rules	• BR-1: Repetitions are counted automatically using validated movement patterns. • BR-2: Exercise performance is evaluated against physiotherapist-defined ROM targets.

Assumptions	• A-1: Patient performs the exercise according to displayed instructions.
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Table 9 UC-9: View Live Digital Twin

Use Case ID	UC-9
Use Case Name	View Live Digital Twin
Actors	Primary Actor: Patient Secondary Actor: AI Analysis Engine, Wearable Sensor Armband
Description	This use case displays a real-time 3D Digital Twin that mirrors the patient's arm movements during rehabilitation. The Digital Twin visualizes joint motion, muscle activation levels, and movement deviations using color-coded indicators.
Trigger	The patient begins a rehabilitation session.
Preconditions	• PRE-1: Rehabilitation session must be active. • PRE-2: Wearable device must be connected. • PRE-3: Sensor data must be streaming successfully.
Postconditions	• POST-1: Digital Twin continuously reflects patient movements. • POST-2: Movement visualization updates in real time.
Normal Flow	1. The rehabilitation session starts. 2. The system receives live IMU and EMG data. 3. The Digital Twin synchronizes with patient movement. 4. Joint movements are rendered in real time. 5. Muscle activation is displayed using color indicators. 6. The patient observes movement quality while exercising.

Alternative Flows	1.1 Visualization Settings1. Patient zooms, rotates, or changes the viewing angle of the Digital Twin. 1.2 Replay Movement1. Patient replays recent movement in slow motion for review.
Exceptions	• E-1: 3D rendering failure. • E-2: Sensor synchronization delay. • E-3: Mobile device performance limitations.
Business Rules	• BR-1: Visualization must refresh continuously throughout the session. • BR-2: Muscle activation colors follow predefined intensity levels (Blue, Green, Orange, Red).
Assumptions	• A-1: Mobile device supports 3D rendering efficiently.

Table 10 UC-10: Receive Real-Time Feedback

Use Case ID	UC-10
Use Case Name	Receive Real-Time Feedback
Actors	Primary Actor: Patient Secondary Actor: AI Analysis Engine
Description	This use-case provides immediate corrective feedback during rehabilitation exercises. The AI continuously analyzes movement quality, muscle activation, ROM, and fatigue levels to generate visual, audio, and safety alerts whenever abnormal movement patterns or excessive fatigue are detected.
Trigger	The AI Analysis Engine detects movement deviations or fatigue during an active rehabilitation session.
Preconditions	• PRE-1: Rehabilitation session must be active. • PRE-2: Sensor data must be continuously available.

	• PRE-3: AI movement analysis must be operational.
Postconditions	• POST-1: Corrective feedback is delivered to the patient. • POST-2: Detected anomalies are logged for physiotherapist review. • POST-3: Exercise quality score is updated.
Normal Flow	1. AI continuously analyzes IMU and EMG sensor data. 2. The system detects movement quality and fatigue. 3. If movement is correct, positive feedback is displayed. 4. If abnormal movement is detected, corrective guidance is generated. 5. Visual indicators appear on the Digital Twin. 6. Audio instructions notify the patient of necessary corrections. 7. Feedback history is stored for physiotherapist review.
Alternative Flows	1.1 Minor Deviation1. The system displays a gentle visual correction only. 1.2 Severe Fatigue Detected1. The system recommends pausing the rehabilitation session.2. The physiotherapist is notified through the dashboard.
Exceptions	• E-1: AI inference service unavailable. • E-2: Sensor data becomes unreliable. • E-3: Real-time communication delay.
Business Rules	• BR-1: Feedback must be generated only from validated AI analysis results. • BR-2: Safety alerts take priority over all other notifications. • BR-3: All detected movement anomalies are stored for future clinical assessment.
Assumptions	• A-1: AI models have been trained using validated rehabilitation datasets. • A-2: Sensor readings accurately represent patient movement and muscle activity.

Table 11 UC-11: View Progress

Use Case ID	UC-11
Use Case Name	View Progress
Actors	Primary Actor: Patient
Description	This use-case allows the patient to monitor rehabilitation progress through detailed performance metrics, including completed sessions, range of motion (ROM), repetition statistics, recovery level, achievement badges, and overall rehabilitation progress. The system presents historical and current performance data to motivate the patient and support continuous recovery.
Trigger	The patient selects View Progress from the dashboard.
Preconditions	• PRE-1: Patient must be logged into the system. • PRE-2: At least one rehabilitation session must have been completed. • PRE-3: Session data must be synchronized with the cloud database.
Postconditions	• POST-1: Patient progress report is displayed. • POST-2: Latest rehabilitation statistics are presented to the patient.
Normal Flow	1. The patient selects View Progress. 2. The system retrieves rehabilitation history from the database. 3. The system calculates updated performance metrics. 4. Progress charts, ROM improvements, completed sessions, and recovery level are displayed. 5. Achievement badges and exercise streaks are updated. 6. The patient reviews overall rehabilitation progress.

Alternative Flows	1.1 No Session History1. The patient has not completed any rehabilitation session.2. The system displays "No progress data available." 1.2 Offline Mode1. Cloud synchronization is unavailable.2. The system displays the latest locally stored progress data.
Exceptions	• E-1: Database retrieval failure. • E-2: Cloud synchronization error. • E-3: Progress calculation service unavailable.
Business Rules	• BR-1: Progress statistics are generated only from completed rehabilitation sessions. • BR-2: Recovery percentage is calculated using physiotherapist-defined ROM targets. • BR-3: Achievement badges are awarded according to predefined rehabilitation milestones.
Assumptions	• A-1: Rehabilitation session data has been successfully stored. • A-2: Progress tracking algorithms are functioning correctly.

Table 12 UC-12: Manage Exercise Database

Use Case ID	UC-12
Use Case Name	Manage Exercise Database
Actors	Primary Actor: Physiotherapist
Description	This use-case allows physiotherapists to manage the clinic's rehabilitation exercise database. They can add new exercises, edit existing exercise information, update rehabilitation instructions, or remove outdated exercises. These exercises are later used to create personalized rehabilitation plans for patients.

Trigger	The physiotherapist selects Manage Exercise Database from the dashboard.
Preconditions	• PRE-1: Physiotherapist must be logged in. • PRE-2: Physiotherapist must have permission to manage the clinic's exercise database. • PRE-3: Exercise database must be available.
Postconditions	• POST-1: Exercise database is successfully updated. • POST-2: Updated exercises become available for patient assignment.
Normal Flow	1. The physiotherapist opens the Exercise Database module. 2. The system displays existing rehabilitation exercises. 3. The physiotherapist chooses to add, edit, or delete an exercise. 4. Exercise details including description, repetitions, ROM recommendations, and instructions are entered or modified. 5. The system validates the entered information. 6. The updated exercise information is saved. 7. The refreshed exercise database is displayed.
Alternative Flows	1.1 Duplicate Exercise 1. The system detects an exercise with the same name. 2. The physiotherapist is prompted to modify the existing exercise instead. 1.2 Cancel Changes 1. The physiotherapist cancels the operation. 2. No modifications are saved.
Exceptions	• E-1: Database connection failure. • E-2: Server error while saving exercise data.
Business Rules	• BR-1: Only authorized physiotherapists may modify exercise records. • BR-2: Exercise names must be unique within the clinic database. • BR-3: Deleted exercises cannot be assigned to new rehabilitation plans.

Assumptions	• A-1: Physiotherapist has sufficient clinical knowledge to create rehabilitation exercises.
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Table 13 UC-13: Assign Exercise Sessions

Use Case ID	UC-13
Use Case Name	Assign Exercise Sessions
Actors	Primary Actor: Physiotherapist Secondary Actor: Clinical Exercise Database, Patient
Description	This use-case enables the physiotherapist to assign personalized rehabilitation exercise sessions to a patient by selecting approved exercises, configuring repetitions, sets, session frequency, and rehabilitation duration according to the patient's condition.
Trigger	The physiotherapist selects Assign Exercise Sessions for a patient.
Preconditions	• PRE-1: Physiotherapist must be logged in. • PRE-2: Patient must be registered and assigned to the physiotherapist. • PRE-3: Exercise database must contain approved rehabilitation exercises.
Postconditions	• POST-1: Personalized rehabilitation plan is assigned to the patient. • POST-2: Patient receives the updated exercise schedule.
Normal Flow	1. The physiotherapist selects a patient. 2. The system displays the available rehabilitation exercises. 3. The physiotherapist selects appropriate exercises. 4. Repetitions, sets, frequency, and rehabilitation duration are configured. 5. The system validates the rehabilitation plan. 6. The exercise sessions are assigned to the patient.

	7. The patient is notified of the new rehabilitation plan.
Alternative Flows	1.1 Modify Existing Plan1. The patient already has an exercise plan.2. The physiotherapist updates the existing rehabilitation schedule. 1.2 Save as Draft1. The physiotherapist saves the plan without assigning it immediately.
Exceptions	• E-1: Patient record unavailable. • E-2: Database update failure.
Business Rules	• BR-1: Only assigned physiotherapists can create rehabilitation plans for their patients. • BR-2: Rehabilitation sessions must use exercises from the approved clinical database.
Assumptions	• A-1: Patient assessment has already been completed.

Table 14 UC-14: Set ROM Targets

Use Case ID	UC-14
Use Case Name	Set ROM Targets
Actors	Primary Actor: Physiotherapist
Description	This use-case allows the physiotherapist to define individualized Range of Motion (ROM) targets and rehabilitation goals for each patient. These targets are later used by the system to evaluate exercise performance and recovery progress.
Trigger	The physiotherapist selects Set ROM Targets for a patient.
Preconditions	• PRE-1: Physiotherapist must be logged in. • PRE-2: Patient profile must exist.

	• PRE-3: Initial patient assessment must be completed.
Postconditions	• POST-1: Patient-specific ROM targets are stored. • POST-2: AI evaluation uses the updated ROM thresholds during rehabilitation sessions.
Normal Flow	1. The physiotherapist selects a patient. 2. Current joint mobility measurements are displayed. 3. Desired ROM targets are entered. 4. Recovery milestones and rehabilitation goals are configured. 5. The system validates the entered values. 6. ROM targets are saved. 7. Updated targets become active for future rehabilitation sessions.
Alternative Flows	1.1 Update Existing Targets 1. The physiotherapist modifies previously defined ROM values. 2. The updated targets replace previous settings. 1.2 Default Clinical Targets 1. The physiotherapist chooses recommended clinical ROM values. 2. The system automatically applies standard rehabilitation targets.
Exceptions	• E-1: Invalid ROM values entered. • E-2: Database update failure.
Business Rules	• BR-1: ROM targets must remain within clinically acceptable limits. • BR-2: Only the assigned physiotherapist may modify ROM targets.
Assumptions	• A-1: Physiotherapist has completed the patient's clinical evaluation.

Table 15 UC-15: Review Session History

Use Case ID	UC-15
Use Case Name	Review Session History
Actors	Primary Actor: Physiotherapist Secondary Actor: Cloud Database
Description	This use case enables the physiotherapist to review previously completed rehabilitation sessions. The session history includes exercise performance, ROM measurements, muscle activation, AI feedback, fatigue alerts, completed repetitions, and overall patient progress to support clinical decision-making.
Trigger	The physiotherapist selects Review Session History from the patient dashboard.
Preconditions	• PRE-1: Physiotherapist must be logged in. • PRE-2: Patient must have completed at least one rehabilitation session. • PRE-3: Session records must be available in the cloud database.
Postconditions	• POST-1: Session history is displayed successfully. • POST-2: Physiotherapist can use the information to adjust future rehabilitation plans.
Normal Flow	1. The physiotherapist selects a patient. 2. The system retrieves completed rehabilitation sessions. 3. Session summaries are displayed chronologically. 4. The physiotherapist selects a session. 5. Detailed performance metrics, AI analysis, ROM graphs, fatigue records, and movement quality reports are displayed. 6. The physiotherapist reviews the patient's rehabilitation progress.

Alternative Flows	1.1 Filter Sessions1. The physiotherapist filters sessions by date or rehabilitation phase.2. The system displays matching records. 1.2 No Session Records1. The patient has not completed any rehabilitation sessions.2. The system displays an appropriate notification.
Exceptions	• E-1: Database retrieval error. • E-2: Cloud synchronization failure. • E-3: Report generation service unavailable.
Business Rules	• BR-1: Physiotherapists may only access session histories of their assigned patients. • BR-2: Rehabilitation records are stored securely and cannot be modified after completion. • BR-3: All session timestamps must be maintained for clinical auditing.
Assumptions	• A-1: Rehabilitation session data has been synchronized successfully with the cloud database. • A-2: Patient consent has been obtained for clinical monitoring.

Table 16 UC-16:Modify Exercise Plan

Use Case ID	UC-16
Use Case Name	Modify Exercise Plan
Actors	Primary Actor: Physiotherapist Secondary Actor: Patient
Description	This use case allows the physiotherapist to modify an existing rehabilitation exercise plan based on the patient's progress, AI analysis, session history, and clinical assessment. The physiotherapist can update exercises, repetitions, sets, ROM targets, session frequency, and rehabilitation duration to better support the patient's recovery.

Trigger	The physiotherapist selects Modify Exercise Plan for an assigned patient.
Preconditions	• PRE-1: Physiotherapist must be logged in. • PRE-2: Patient must have an existing rehabilitation plan. • PRE-3: Physiotherapist must be assigned to the patient.
Postconditions	• POST-1: Updated rehabilitation plan is saved successfully. • POST-2: Patient receives the revised exercise plan. • POST-3: Future rehabilitation sessions use the updated plan.
Normal Flow	1. The physiotherapist selects a patient. 2. The current rehabilitation plan is displayed. 3. The physiotherapist reviews session history and AI recommendations. 4. Required changes are made to exercises, repetitions, sets, ROM targets, or session frequency. 5. The system validates the updated rehabilitation plan. 6. The revised plan is saved in the database. 7. The patient is notified about the updated rehabilitation program.
Alternative Flows	1.1 Save as Draft 1. The physiotherapist saves modifications without publishing them immediately. 2. The previous exercise plan remains active. 1.2 Cancel Modification 1. The physiotherapist cancels the modification process. 2. No changes are saved.
Exceptions	• E-1: Database update failure. • E-2: Exercise database unavailable. • E-3: Network interruption while saving changes.
Business Rules	• BR-1: Only the assigned physiotherapist may modify a patient's rehabilitation plan.

	• BR-2: Modified plans become effective immediately after successful saving.
Assumptions	• A-1: The physiotherapist has reviewed the patient's rehabilitation progress before making modifications. • A-2: Updated exercises exist in the approved exercise database.

Table 17 UC-17: Add Physiotherapist

Use Case ID	UC-17
Use Case Name	Add Physiotherapist
Actors	Primary Actor: Clinic Administrator
Description	This use-case allows the clinic administrator to register a new physiotherapist within the Inteli-Rehab system. The administrator enters the physiotherapist's professional and personal information, and the system creates a secure account associated with the clinic.
Trigger	The clinic administrator selects Add Physiotherapist from the administration dashboard.
Preconditions	• PRE-1: Clinic administrator must be logged in. • PRE-2: Administrator must have account management privileges.
Postconditions	• POST-1: Physiotherapist account is created successfully. • POST-2: Physiotherapist is linked to the clinic. • POST-3: Login credentials are generated and shared securely.
Normal Flow	1. The administrator selects Add Physiotherapist. 2. The system displays the registration form. 3. The administrator enters physiotherapist details including name, email, contact information, specialization, and license information.

	4. The system validates the entered information. 5. The system creates the physiotherapist account. 6. The physiotherapist is linked with the clinic. 7. Confirmation message is displayed.
Alternative Flows	1.1 Existing Email 1. The entered email is already registered. 2. The system requests another email address. 1.2 Incomplete Information 1. Required information is missing. 2. The administrator is prompted to complete all mandatory fields.
Exceptions	• E-1: Authentication server unavailable. • E-2: Database insertion failure. • E-3: Internet connectivity interruption.
Business Rules	• BR-1: Each physiotherapist must have a unique registered email address. • BR-2: Only clinic administrators can create physiotherapist accounts. • BR-3: Every physiotherapist account must be associated with exactly one clinic.
Assumptions	• A-1: The physiotherapist possesses valid professional credentials. • A-2: The clinic has available capacity to register additional physiotherapists.

Table 18 UC-18: Remove Physiotherapist

Use Case ID	UC-18
Use Case Name	Remove Physiotherapist
Actors	Primary Actor: Clinic Administrator
Description	This use-case allows the clinic administrator to remove a physiotherapist from the clinic. The system deactivates the physiotherapist's account.

Trigger	The clinic administrator selects Remove Physiotherapist.
Preconditions	• PRE-1: Clinic administrator must be logged in. • PRE-2: Physiotherapist account must exist. • PRE-3: Administrator must have account management privileges.
Postconditions	• POST-1: Physiotherapist account is deactivated or removed from the clinic. • POST-2: Existing rehabilitation records remain securely stored. • POST-3: Patients can no longer be assigned to the removed physiotherapist.
Normal Flow	1. The administrator opens the physiotherapist management page. 2. The system displays all physiotherapists. 3. The administrator selects a physiotherapist. 4. The administrator confirms the removal action. 5. The system verifies whether active patients are assigned. 6. The physiotherapist account is deactivated. 7. Confirmation message is displayed.
Alternative Flows	1.1 Active Patients Assigned 1. The physiotherapist still has active patients. 2. The system requests patient reassignment before removal. 1.2 Cancel Removal 1. The administrator cancels the operation. 2. The physiotherapist account remains active.
Exceptions	• E-1: Database update failure. • E-2: Server unavailable. • E-3: Permission validation failure.
Business Rules	• BR-1: Physiotherapists with active patients cannot be removed until reassignment is completed. • BR-2: Only clinic administrators may remove physiotherapist accounts.

Assumptions	• A-1: Patient reassignment procedures are available when required.
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Table 19 UC-19: Manage Physiotherapist Accounts

Use Case ID	UC-19
Use Case Name	Manage Physiotherapist Accounts
Actors	Primary Actor: Clinic Administrator
Description	This use case allows the clinic administrator to manage physiotherapist accounts by viewing account information, updating professional details, and maintaining clinic staff records. It ensures that physiotherapist information remains accurate and up to date.
Trigger	The clinic administrator selects Manage Physiotherapist Accounts from the administration dashboard.
Preconditions	• PRE-1: Clinic administrator must be logged in. • PRE-2: Administrator must possess account management privileges. • PRE-3: Physiotherapist accounts must exist in the system.
Postconditions	• POST-1: Physiotherapist account information is added successfully. • POST-2: Account status changes are reflected throughout the system. • POST-3: Updated information becomes immediately available to authorized users.
Normal Flow	1. The administrator opens Manage Physiotherapist Accounts. 2. The system displays all registered physiotherapists. 3. The administrator selects an account. 4. The administrator views profile information, specialization, contact details, account status.

	5. The system validates the entered information, for adding a new physiotherapist. 6. Changes are saved to the database.
Alternative Flows	1.1 Deactivate Account1. The administrator deactivates the physiotherapist account.2. The physiotherapist is prevented from accessing the system until reactivated.
Exceptions	• E-1: Database update failure. • E-2: Authentication service unavailable. • E-3: Network timeout while saving changes.
Business Rules	• BR-1: Only clinic administrators are authorized to manage physiotherapist accounts. • BR-2: Deactivated physiotherapists cannot log into the Inteli-Rehab system.
Assumptions	• A-1: Authentication and database services are functioning correctly. • A-2: The clinic administrator follows organizational account management policies.

Functional Requirements

This section presents the core functional requirements of the **INTELI-REHAB** system. Each requirement describes a major function that the system must perform to support AI-powered upper-limb rehabilitation using wearable sensors, Digital Twin visualization, and physiotherapist-assisted monitoring.

1. Patient Registration and Login

Table 20 FR-1 Patient Registration and Login

Identifier	FR-1
Title	Patient Registration and Login
Requirement	The system shall allow patients to register and securely log in under an assigned physiotherapist from a selected clinic or hospital using authenticated credentials.

Source	UC-1 (Register Account), UC-2 (Login)
Rationale	To provide secure access to rehabilitation services while ensuring every patient is associated with an authorized physiotherapist.
Business Rule	BR-1: Each patient account must be linked to one registered physiotherapist and one clinic.
Dependencies	FR-2 (Role-Based Authentication), FR-3 (Patient Profile Management)
Priority	High

2. Role-Based Authentication

Table 21FR-2: Role-Based Authentication

Identifier	FR-2
Title	Role-Based Authentication
Requirement	The system shall authenticate physiotherapists and clinic administrators using secure login credentials and provide access according to their assigned roles and permissions.
Source	UC-2 (Login), UC-17 (Add Physiotherapist), UC-19 (Manage Physiotherapist Accounts)
Rationale	To protect sensitive rehabilitation data and ensure users only access authorized system functions.
Business Rule	BR-2: Access permissions shall be granted according to predefined user roles (Patient, Physiotherapist, Clinic Administrator).
Dependencies	FR-1 (Patient Registration and Login)
Priority	High

3. Patient Rehabilitation Profile Management

Table 22 FR-3: Patient Rehabilitation Profile Management

Identifier	FR-3
Title	Patient Rehabilitation Profile Management
Requirement	The system shall store and manage patient rehabilitation profiles, including injury type, affected joint, recovery stage, medical information, and assigned physiotherapist.
Source	UC-4 (Manage Patient Profile)
Rationale	To maintain complete rehabilitation records required for personalized treatment planning and monitoring.
Business Rule	BR-3: Only the assigned physiotherapist and authorized clinic administrator may access or modify patient rehabilitation profiles.
Dependencies	FR-1 (Patient Registration and Login), FR-2 (Role-Based Authentication)
Priority	High

4. Rehabilitation Exercise Database

Table 23 FR-4: Rehabilitation Exercise Database

Identifier	FR-4
Title	Rehabilitation Exercise Database
Requirement	The system shall maintain a rehabilitation exercise database containing exercise descriptions, target joint movements, muscle activation information, rehabilitation instructions, and clinical guidelines.
Source	UC-12 (Manage Exercise Database)
Rationale	To provide physiotherapists with standardized rehabilitation exercises for creating personalized treatment plans.

Business Rule	BR-4: Only authorized physiotherapists may create, modify, or delete rehabilitation exercises.
Dependencies	FR-2 (Role-Based Authentication)
Priority	High

5. Personalized Exercise Plan Management

Table 24 FR-5: Personalized Exercise Plan Management

Identifier	FR-5
Title	Personalized Exercise Plan Management
Requirement	The system shall enable physiotherapists to create, assign, modify, and manage personalized rehabilitation exercise plans, including repetitions, sets, session duration, recovery intervals, exercise frequency, and target Range of Motion (ROM).
Source	UC-13 (Assign Exercise Sessions), UC-14 (Set ROM Targets), UC-16 (Modify Exercise Plan)
Rationale	To provide individualized rehabilitation programs that match each patient's injury condition and recovery progress.
Business Rule	BR-5: Exercise plans may only be created or modified by the patient's assigned physiotherapist, and all modifications shall be recorded for clinical auditing.
Dependencies	FR-3 (Patient Rehabilitation Profile Management), FR-4 (Rehabilitation Exercise Database), FR-6 (Exercise Plan Version Management)
Priority	High

6. Exercise Plan Version Management

Table 25 FR-6: Exercise Plan Version Management

Identifier	FR-6
Title	Exercise Plan Version Management
Requirement	The system shall maintain version history for all rehabilitation exercise plans and protocol modifications, allowing physiotherapists to review previous versions and track treatment changes.
Source	UC-16 (Modify Exercise Plan)
Rationale	To preserve rehabilitation history, support clinical decision-making, and provide traceability of treatment modifications.
Business Rule	BR-6: Every modification to an exercise plan shall be automatically timestamped and linked to the responsible physiotherapist.
Dependencies	FR-5 (Personalized Exercise Plan Management)
Priority	Medium

7. Wearable Device Connectivity

Table 26 FR-7: Wearable Device Connectivity

Identifier	FR-7
Title	Wearable Device Connectivity
Requirement	The system shall establish and maintain Bluetooth Low Energy (BLE) communication between the wearable rehabilitation device and the mobile application before and during rehabilitation sessions.
Source	UC-5 (Connect Wearable Device), UC-7 (Start Rehabilitation Session)
Rationale	To enable reliable communication between wearable sensors and the rehabilitation application for continuous data acquisition.

Business Rule	BR-7: Rehabilitation sessions shall not begin unless the wearable device is successfully connected and verified.
Dependencies	FR-9 (Sensor Calibration and Motion Processing)
Priority	High

8. Sensor Data Acquisition

Table 27 FR-8: Sensor Data Acquisition

Identifier	FR-8
Title	Sensor Data Acquisition
Requirement	The system shall continuously collect IMU and EMG sensor data during rehabilitation exercises and automatically reconnect the wearable device if Bluetooth communication is interrupted.
Source	UC-7 (Start Rehabilitation Session), UC-8 (Perform Exercise)
Rationale	To ensure uninterrupted acquisition of movement and muscle activity data for accurate rehabilitation analysis.
Business Rule	BR-8: Sensor data shall be continuously monitored throughout every active rehabilitation session, and automatic reconnection shall be attempted after communication loss.
Dependencies	FR-7 (Wearable Device Connectivity)
Priority	High

9. Sensor Calibration and Motion Processing

Table 28 FR-9: Sensor Calibration and Motion Processing

Identifier	FR-9
Title	Sensor Calibration and Motion Processing

Requirement	The system shall guide patients through sensor calibration, establish biometric baselines, preprocess IMU and EMG signals, and estimate upper-limb orientation, joint angles, and movement parameters before rehabilitation sessions begin.
Source	UC-5 (Calibrate Sensors), UC-7 (Start Rehabilitation Session)
Rationale	To improve the accuracy of movement tracking, muscle activity analysis, and rehabilitation assessment.
Business Rule	BR-9: Rehabilitation sessions shall only start after successful sensor calibration and biometric baseline establishment.
Dependencies	FR-7 (Wearable Device Connectivity), FR-8 (Sensor Data Acquisition), FR-10 (Rehabilitation Monitoring)
Priority	High

10.Rehabilitation Monitoring

Table 29 FR-10:Rehabilitation Monitoring

Identifier	FR-10
Title	Rehabilitation Monitoring
Requirement	The system shall monitor shoulder and elbow movements in real time by calculating joint angles, repetition counts, rest intervals, and Range of Motion (ROM) throughout rehabilitation exercises.
Source	UC-8 (Perform Exercise), UC-10 (Receive Real-Time Feedback)
Rationale	To evaluate rehabilitation performance accurately and provide reliable movement measurements for patient recovery assessment.
Business Rule	BR-10: Joint angles, repetitions, and ROM measurements shall be calculated continuously throughout each rehabilitation session and compared with physiotherapist-defined rehabilitation targets.

Dependencies	FR-8 (Sensor Data Acquisition), FR-9 (Sensor Calibration and Motion Processing), FR-11 (Progress Monitoring and Reporting)
Priority	High

11. Progress Monitoring and Reporting

Table 30 FR-11: Progress Monitoring and Reporting

Identifier	FR-11
Title	Progress Monitoring and Reporting
Requirement	The system shall maintain rehabilitation history and generate progress reports, including Range of Motion (ROM) trends, exercise performance statistics, rehabilitation progress, and recovery history for both patients and physiotherapists.
Source	UC-9 (View Progress Report), UC-11 (View Patient Progress)
Rationale	To enable continuous monitoring of rehabilitation progress and assist physiotherapists in evaluating patient recovery over time.
Business Rule	BR-11: Progress reports shall be generated only from validated rehabilitation session data and be accessible only to authorized users.
Dependencies	FR-10 (Rehabilitation Monitoring), FR-16 (Physiotherapist Dashboard)
Priority	High

12. Digital Twin Visualization

Table 31 FR-12: Digital Twin Visualization

Identifier	FR-12
Title	Digital Twin Visualization

Requirement	The system shall generate a real-time three-dimensional (3D) Digital Twin of the patient's upper limb, displaying shoulder and elbow movements, muscle activation levels, and posture deviations during rehabilitation exercises.
Source	UC-8 (Perform Exercise), UC-10 (Receive Real-Time Feedback)
Rationale	To provide patients and physiotherapists with an intuitive visual representation of rehabilitation performance and movement quality.
Business Rule	BR-12: The Digital Twin shall update continuously using real-time sensor data throughout the rehabilitation session.
Dependencies	FR-8 (Sensor Data Acquisition), FR-9 (Sensor Calibration and Motion Processing), FR-10 (Rehabilitation Monitoring)
Priority	High

13.AI Movement Assessment

Table 32 FR-13: Movement Assessment

Identifier	FR-13
Title	AI Movement Assessment
Requirement	The system shall analyze rehabilitation movements using Artificial Intelligence (AI) to detect abnormal movement patterns, incorrect posture, movement compensations, and deviations from prescribed rehabilitation exercises.
Source	UC-10 (Receive Real-Time Feedback), UC-15 (Review Rehabilitation Session)
Rationale	To improve rehabilitation quality by automatically identifying movement errors that may affect patient recovery.
Business Rule	BR-13: AI analysis shall compare patient movements against physiotherapist-approved rehabilitation targets before generating corrective assessments.
Dependencies	FR-10 (Rehabilitation Monitoring), FR-12 (Digital Twin Visualization), FR-14 (Corrective Feedback System)

Priority	High
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14. Corrective Feedback System

Table 33 FR-14 Corrective Feedback System

Identifier	FR-14
Title	Corrective Feedback System
Requirement	The system shall provide real-time corrective feedback through visual indicators, voice guidance, and rehabilitation recommendations while recording detected movement anomalies for physiotherapist review.
Source	UC-10 (Receive Real-Time Feedback), UC-15 (Review Rehabilitation Session)
Rationale	To assist patients in performing rehabilitation exercises correctly and reduce the risk of improper movements during recovery.
Business Rule	BR-14: Corrective alerts shall only be generated when movement deviations exceed predefined rehabilitation thresholds established by the physiotherapist.
Dependencies	FR-13 (AI Movement Assessment), FR-15 (Fatigue Monitoring)
Priority	High

15. Fatigue Monitoring

Table 34 FR-15: Fatigue Monitoring

Identifier	FR-15
Title	Fatigue Monitoring
Requirement	The system shall monitor muscle fatigue using EMG signals and provide progressive safety alerts, including visual notifications, voice warnings, and rehabilitation session pause recommendations when fatigue thresholds are exceeded.

Source	UC-10 (Receive Real-Time Feedback), UC-15 (Review Rehabilitation Session)
Rationale	To ensure patient safety by detecting excessive muscle fatigue and preventing overexertion during rehabilitation exercises.
Business Rule	BR-15: The system shall issue graded fatigue warnings based on patient-specific calibration baselines established before rehabilitation sessions.
Dependencies	FR-9 (Sensor Calibration and Motion Processing), FR-13 (AI Movement Assessment), FR-14 (Corrective Feedback System)
Priority	High

16. Physiotherapist Dashboard

Table 35 FR-16: Physiotherapist Dashboard

Identifier	FR-16
Title	Physiotherapist Dashboard
Requirement	The system shall provide a physiotherapist dashboard for monitoring patient adherence, rehabilitation performance, Range of Motion (ROM) progress, AI-generated alerts, rehabilitation history, and Digital Twin session summaries.
Source	UC-11 (View Patient Progress), UC-15 (Review Rehabilitation Session)
Rationale	To provide physiotherapists with a centralized platform for monitoring patient rehabilitation progress and making informed clinical decisions.
Business Rule	BR-16: Physiotherapists shall only access rehabilitation data of patients assigned to them.
Dependencies	FR-11 (Progress Monitoring and Reporting), FR-13 (AI Movement Assessment), FR-19 (Cloud Synchronization and Notifications)
Priority	High

17.Rehabilitation Plan Management

Table 36 FR-17: Rehabilitation Plan Management

Identifier	FR-17
Title	Rehabilitation Plan Management
Requirement	The system shall allow physiotherapists to prescribe rehabilitation exercises, modify exercise plans, approve Range of Motion (ROM) targets, review rehabilitation session records, and adjust treatment plans based on patient progress.
Source	UC-13 (Assign Exercise Sessions), UC-14 (Set ROM Targets), UC-16 (Modify Exercise Plan)
Rationale	To enable physiotherapists to personalize rehabilitation programs according to each patient's recovery status and clinical assessment.
Business Rule	BR-17: Only the assigned physiotherapist may prescribe or modify a patient's rehabilitation plan, and all modifications shall be recorded in the system audit log.
Dependencies	FR-5 (Personalized Exercise Plan Management), FR-6 (Exercise Plan Version Management), FR-16 (Physiotherapist Dashboard)
Priority	High

18.Clinic Administration

Table 37 FR-8 Clinic Administration

Identifier	FR-18
Title	Clinic Administration
Requirement	The system shall provide a clinic administration panel for managing physiotherapist accounts, assigning physiotherapists to patients, activating or deactivating accounts, and maintaining clinic staff records.

Source	UC-17 (Add Physiotherapist), UC-18 (Remove Physiotherapist), UC-19 (Manage Physiotherapist Accounts)
Rationale	To enable efficient management of physiotherapist accounts and ensure proper allocation of rehabilitation services within the clinic.
Business Rule	BR-18: Only clinic administrators shall be authorized to create, update, deactivate, or remove physiotherapist accounts. Historical rehabilitation records shall remain preserved after account deactivation.
Dependencies	FR-2 (Role-Based Authentication), FR-16 (Physiotherapist Dashboard)
Priority	Medium

19.Cloud Synchronization and Notifications

Table 38 FR-19: Cloud Synchronization and Notifications

Identifier	FR-19
Title	Cloud Synchronization and Notifications
Requirement	The system shall synchronize rehabilitation data with the cloud database and notify physiotherapists of significant patient events, including abnormal movement patterns, missed rehabilitation sessions, critical muscle fatigue alerts, and rehabilitation progress updates.
Source	UC-15 (Review Rehabilitation Session), UC-19 (Manage Physiotherapist Accounts)
Rationale	To ensure rehabilitation data remains available across the system while enabling timely intervention through automated notifications and cloud-based synchronization.
Business Rule	BR-19: Rehabilitation data shall be synchronized securely with the cloud, and notifications shall only be delivered to authorized physiotherapists and clinic administrators responsible for the patient.
Dependencies	FR-11 (Progress Monitoring and Reporting), FR-13 (AI Movement Assessment), FR-15 (Fatigue Monitoring), FR-16 (Physiotherapist Dashboard)

Priority	High
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Non-Functional Requirements

This section specifies the quality attributes of the **INTELI-REHAB** system, including usability, performance, reliability, security, scalability, maintainability, and portability. These requirements ensure that the rehabilitation platform provides a secure, responsive, reliable, and user-friendly environment for patients, physiotherapists, and clinic administrators.

1. Usability

USE-1: The system shall provide an intuitive and responsive mobile and web interface that enables patients to start a rehabilitation session within **three user interactions** after logging in.

USE-2: The system shall display clear visual indicators, icons, and color-coded feedback for exercise status, Digital Twin visualization, muscle activation, posture correction, and rehabilitation progress.

USE-3: The system shall provide real-time visual and voice guidance during rehabilitation exercises to help patients perform movements correctly with minimal physiotherapist intervention.

USE-4: The physiotherapist dashboard shall present rehabilitation progress, ROM trends, alerts, and patient performance using easy-to-understand charts and graphical reports.

2. Performance

PER-1: The system shall establish a Bluetooth Low Energy (BLE) connection with the wearable rehabilitation device within **5 seconds** under normal operating conditions.

PER-2: The system shall process IMU and EMG sensor data and update the Digital Twin visualization with an average latency of **less than 10 milliseconds**.

PER-3: The AI movement analysis module shall detect incorrect posture and movement abnormalities within **3 second** after receiving sensor data.

PER-4: The physiotherapist dashboard shall load patient rehabilitation history and progress reports within **3 seconds** for datasets containing up to **100 rehabilitation sessions**.

PER-5: The backend shall respond to standard API requests, including authentication, exercise plan management, and rehabilitation data retrieval, within **2 seconds** under normal operating conditions.

3. Reliability and Availability

REL-1: The system shall maintain at least **99% availability** during normal operation, excluding scheduled maintenance.

REL-2: In case of Bluetooth disconnection, the system shall automatically attempt to reconnect the wearable device without terminating the rehabilitation session.

REL-3: Rehabilitation session data shall be stored reliably to prevent loss during unexpected application or network failures.

REL-4: The system shall display meaningful error messages and allow users to retry failed operations caused by temporary communication or server failures.

REL-5: Patient profiles, rehabilitation history, and exercise plans shall be backed up automatically every **24 hours**.

4. Security

SEC-1: The system shall implement Role-Based Access Control (RBAC) to restrict system functionality according to Patient, Physiotherapist, and Clinic Administrator roles.

SEC-2: User passwords shall be securely hashed before being stored in the database.

SEC-3: User sessions shall automatically expire after **30 minutes** of inactivity to prevent unauthorized access.

5. Scalability

SCA-1: The system architecture shall support adding multiple clinics, physiotherapists, and patients without requiring significant software modifications.

SCA-2: The cloud database shall support efficient storage and retrieval of increasing rehabilitation records through optimized indexing techniques.

SCA-3: The backend shall support concurrent rehabilitation sessions from multiple patients without significant degradation in performance.

SCA-4: The system architecture shall support future integration with additional wearable sensors and rehabilitation modules.

6. Maintainability

MAI-1: The system shall be developed using a modular architecture, allowing independent updates to mobile application, AI processing, Digital Twin visualization, physiotherapist dashboard, and cloud services.

MAI-2: The project source code shall follow standard coding conventions and include sufficient inline documentation for future maintenance.

MAI-3: The system shall support unit testing and integration testing to simplify software verification and regression testing.

MAI-4: Software updates and bug fixes shall be managed using a Git-based version control system to ensure maintainability and traceability.

7. Portability

POR-1: The mobile application shall support Android devices running Android 10 or later.

POR-2: The physiotherapist web dashboard shall be compatible with the latest versions of Google Chrome, Microsoft Edge.

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